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| **Fiscal Year:** | FY 2020 | **Task Last Updated:** | FY 08/03/2020 |
| **PI Name:** | Javanmard, Mehdi Ph.D. | | |
| **Project Title:** | The Cytotracker: Point-of-use Complete Blood Count | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | TRISH--TRISH | | |
| **Joint Agency Name:** | | **TechPort:** | Yes |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 08540 | **Congressional District:** | 12 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | TRISH--Industry |
| **Start Date:** | 05/01/2019 | **End Date:** | 04/30/2020 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | TRISH |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX16AO69A-IND0004 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | Industry Project<br>The objective of this proposal is to build and characterize an ultra-compact all-electronic cytometer capable of quantifying key cellular components in a complete blood cell count (CBC) including: red blood cells (RBC), white blood cells (WBC), platelets, hemoglobin (hgb) concentration, and hematocrit levels. We propose development of an analyzer that is at least one order of magnitude lower in cost than the state of the art analyzers and requires only 5 µl of blood. While RBCs, WBCs, and platelets can be measured electrically, a primary issue hindering impedance cytometry from providing accurate cell counts is the wide signal variation observed due to device-to-device variation, drift due to temperature fluctuations, and off-axis cell translocation in microfluidic channels. Hgb and hematocrit are typically measured based on optical scattering, thus making difficult the realization of a fully electronic platform that can measure the five key parameters mentioned above. Because the combined use of electronics and optics in a single | | |

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| **Task Description:** | cytometry platform makes miniaturization to a pocket-sized platform difficult, an innovative approach is proposed to improving accuracy of impedance cytometers to measuring the five key parameters above, without requiring additional optics, enabling realization of a pocket-sized platform. We propose use of a multi-frequency impedance cytometer (MFIC) embedded with multiple sensors to enable adaptive self-calibration in conjunction with machine learning to discriminate between RBCs, WBCs, platelets, while estimating hgb and hematocrit levels with less than 5 µl of blood.<br><br>To develop this product, the following objectives are proposed: Objective 1) Integrated microfluidic biosensor fabrication and testing. Objective 2) Developing datadriven software for accurate classification of blood cell types, hemoglobin, and hematocrit using impedance cytometry data. Objective 3) Ultra-compact pocket-sized hardware and smartphone application development. Objective 4) Benchmarking and testing of device accuracy against Food and Drug Administration (FDA) approved predicate device. The result of this project will be a working prototype of the proposed analyzer.<br><br>The intellectual merits are as following: 1) Engineering an all-electronic data-driven bio-sensing platform capable of label-free cell classification. 2) The development of a portable wireless instrument that harnesses machine-learning to accurately classify cells and display results rapidly. 3) The development of ultra-compact modules, which can be applied to a broad range of assays beyond complete blood count analysis. This project will lay the groundwork for development of a commercial product that can improve access to healthcare not only for astronauts in space but also broader society. We will develop an analyzer that is i) at least one order of magnitude lower in cost than the state of the art analyzers, ii) easily wearable/portable (pocket-sized) and lightweight, iii) rapid (provides results within minutes), and iv) easy-to-use and requires only a finger prick of blood. The accessible cost and portability of the proposed inexpensive lightweight user-friendly cytometer to perform automated CBC without the need for professional intervention has a multitude of applications that can significantly impact primary care practice in the United States and globally. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The project has many impacts and applications for space health, including but not limited to: 1) Rapid assessment of hct/hgb can diagnose anemia and dehydration in astronauts. 2) Quantification of RBC levels can diagnose sleep apnea in astronauts. 3) Rapid quantification of WBC levels (neutrophils and lymphocytes) can monitor effect of radiation on the immune system. 4) Quantification of platelets can diagnose various platelet related conditions (acute and chronic) for astronauts. There are many applications and high impact use cases for a rapid portable point-of-need blood cell count (CBC) analyzer for terrestrial health applications also including: 1) Testing in the emergency room: Trauma, internal bleeding, infection. 2) Testing on the Field: military. 3) Urgent care clinics. 4) Cancer patients, and even 5) Home based testing. |
| **Task Progress:** | The results of the research project are summarized below:<br>Detection of platelets was a capability that we did not have before beginning this Translational Research Institute for Space Health (TRISH) project. This is challenging with impedance cytometry since platelets are small in size (barely above noise). We were able to tune the parameters of the lock-in-amplifier circuit to have sufficient signal-to-noise ratio (SNR) for platelets. As for red blood cells (RBCs), we were able to achieve detection and quantification within 2 minutes. We rapidly prototyped soft-lithographically fabricated channels. Initial results are encouraging despite the crudeness of the fabrication process. RBC concentration and hematocrit levels were characterized for over 15 samples in triplicate (45 experiments) compared to Food and Drug Administration (FDA) approved predicate device. Before this TRISH project, we did not have capability to detect hematocrit levels. We developed a novel algorithm that utilizes the peak intensity data obtained from counting individual RBCs, computing the volume, summing the volume of all cells, and dividing total volume of RBCs. Our algorithm performs metering of the blood by measuring the peak width, which is equal to the transit time of the cell passing through the cytometer sensor. The transit time is converted to velocity, thus flow rate and volume that has passed the sensor is computed. 25 samples were analyzed for platelet concentration (five different concentrations were measured). An average standard error of 6% was achieved. |
| **Bibliography Type:** | Description: (Last Updated: ) |